Lesson 03 Demo 02

Setting up Kubernetes Cluster on EC2 Instances

Objective: To set up a Kubernetes cluster on Amazon EC2 instances, ensuring seamless communication between a master node and a worker node

Tools required: AWS Management Console

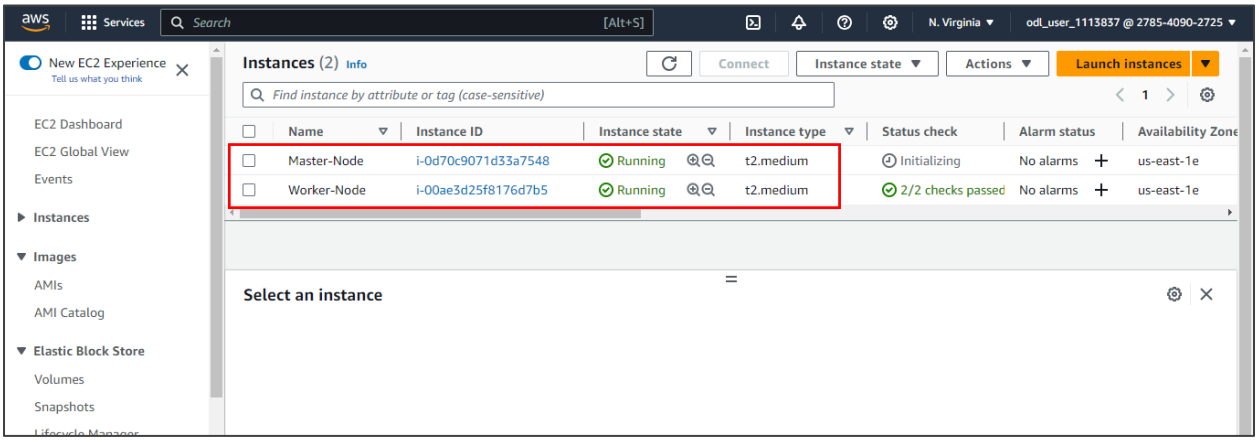
Prerequisites: None

Steps to be followed:

1. Set up the EC2 instances
2. Configure the Kubernetes master node
3. Connect the worker instance to the master node

Step 1: Set up the EC2 instances

1.1 Launch two EC2 instances using the Ubuntu image: one for the master node and one for the worker node. Ensure the instance type is **t2.medium** or higher.



Note: Refer to previous demos on how to launch an EC2 instance

1.2 Install the necessary Kubernetes components using the commands given below:

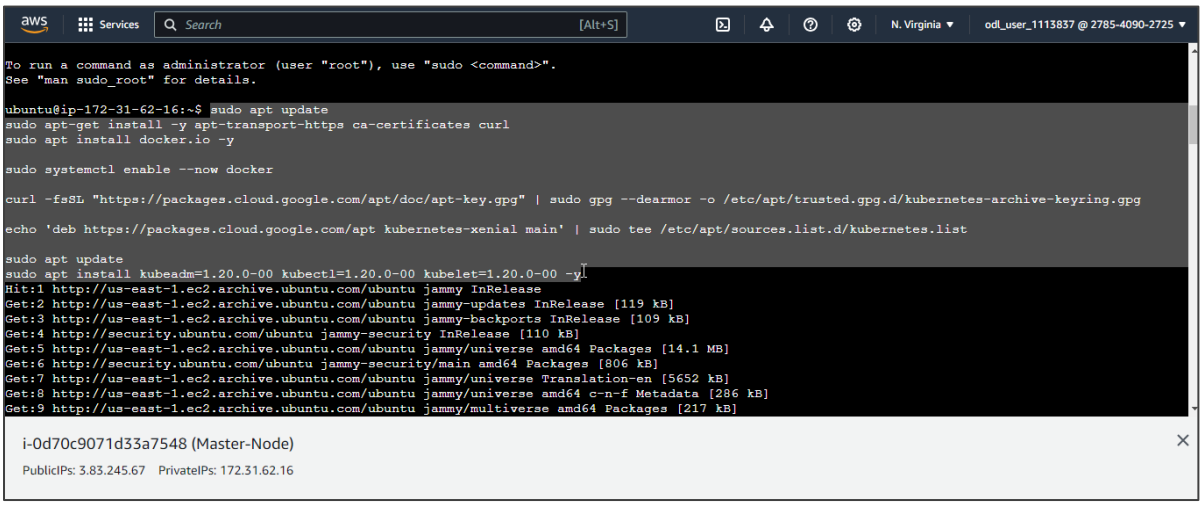
sudo apt update
sudo apt-get install -y apt-transport-https ca-certificates curl
sudo apt install docker.io -y

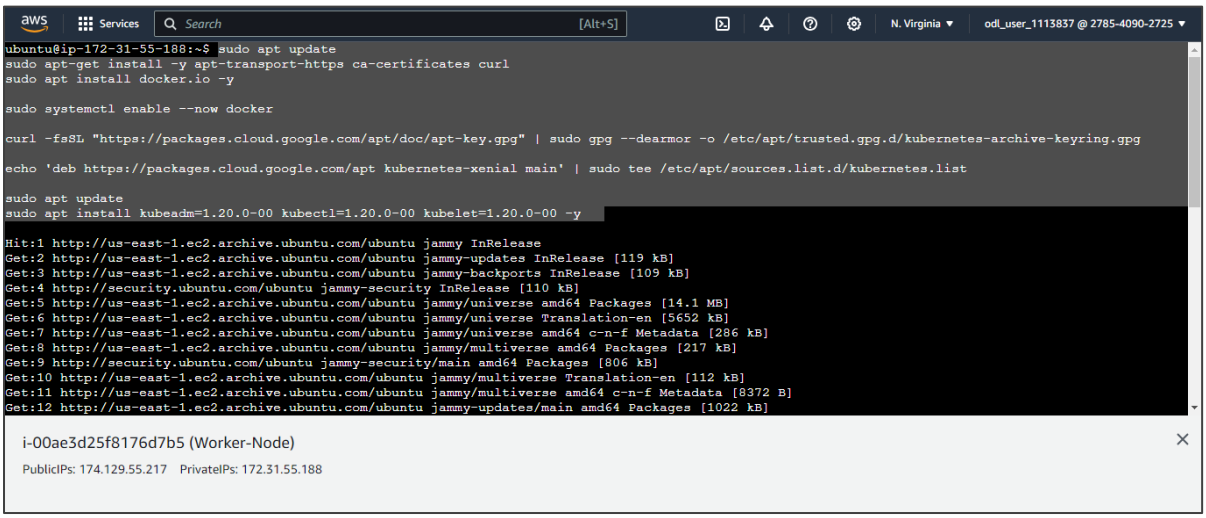
sudo systemctl enable --now docker

curl -fsSL "https://packages.cloud.google.com/apt/doc/apt-key.gpg" | sudo gpg --dearmor -o /etc/apt/trusted.gpg.d/kubernetes-archive-keyring.gpg

echo 'deb https://packages.cloud.google.com/apt kubernetes-xenial main' | sudo tee /etc/apt/sources.list.d/kubernetes.list

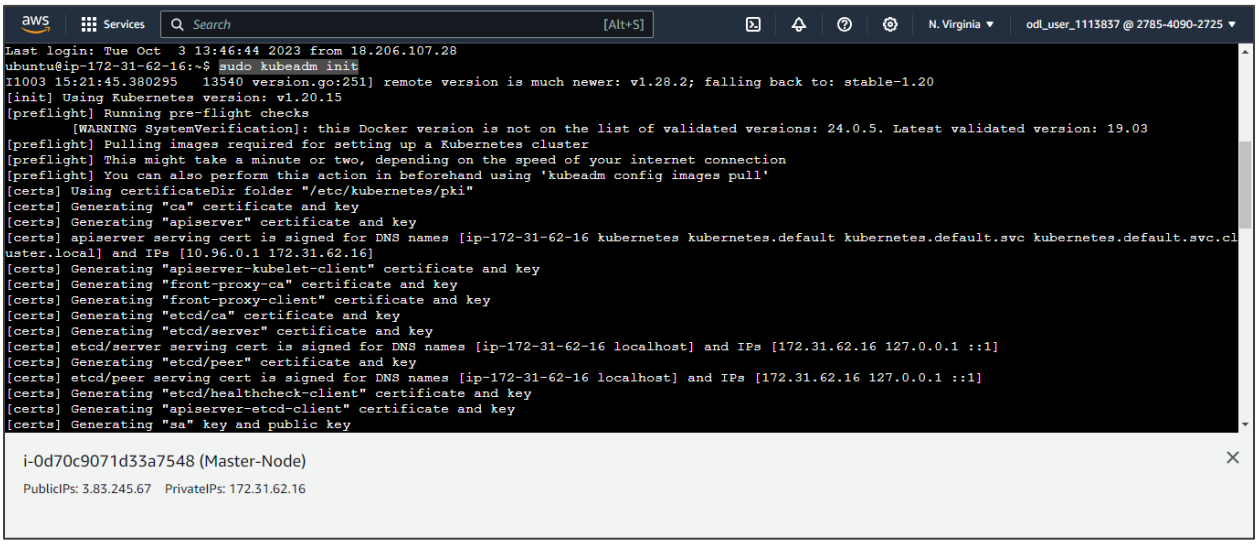
sudo apt update
sudo apt install kubeadm=1.20.0-00 kubectl=1.20.0-00 kubelet=1.20.0-00 -y



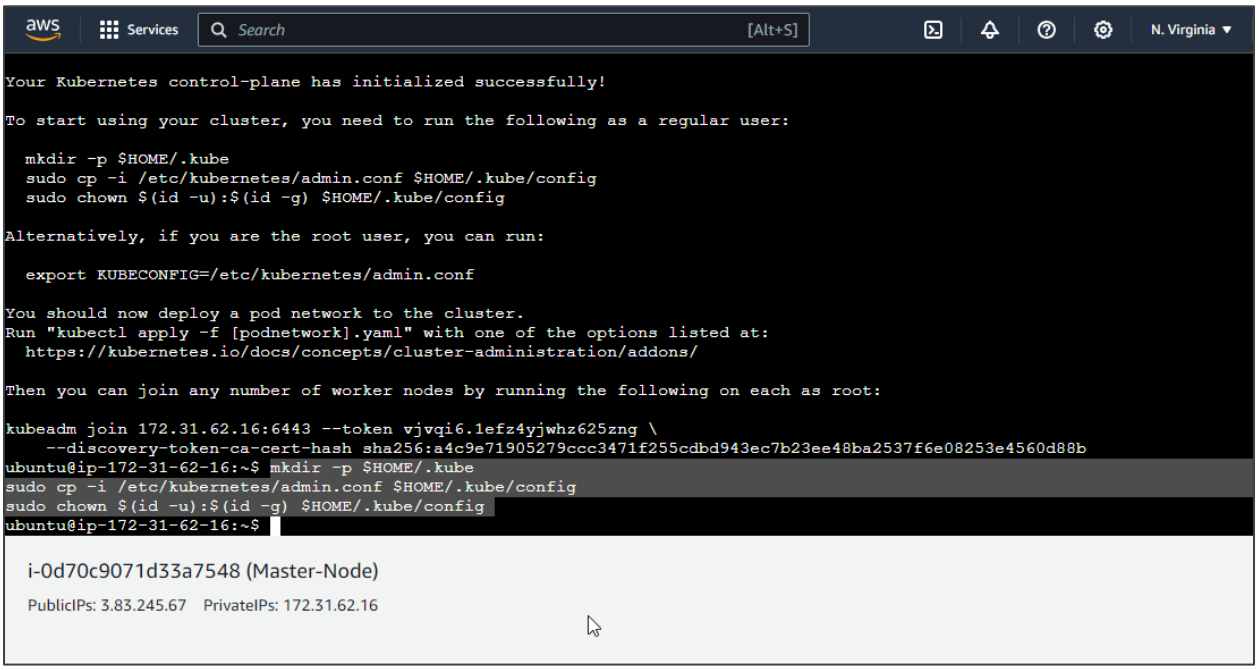


Step 2: Configure the Kubernetes master node

2.1 Run the following command to initialize the master node:
sudo kubeadm init

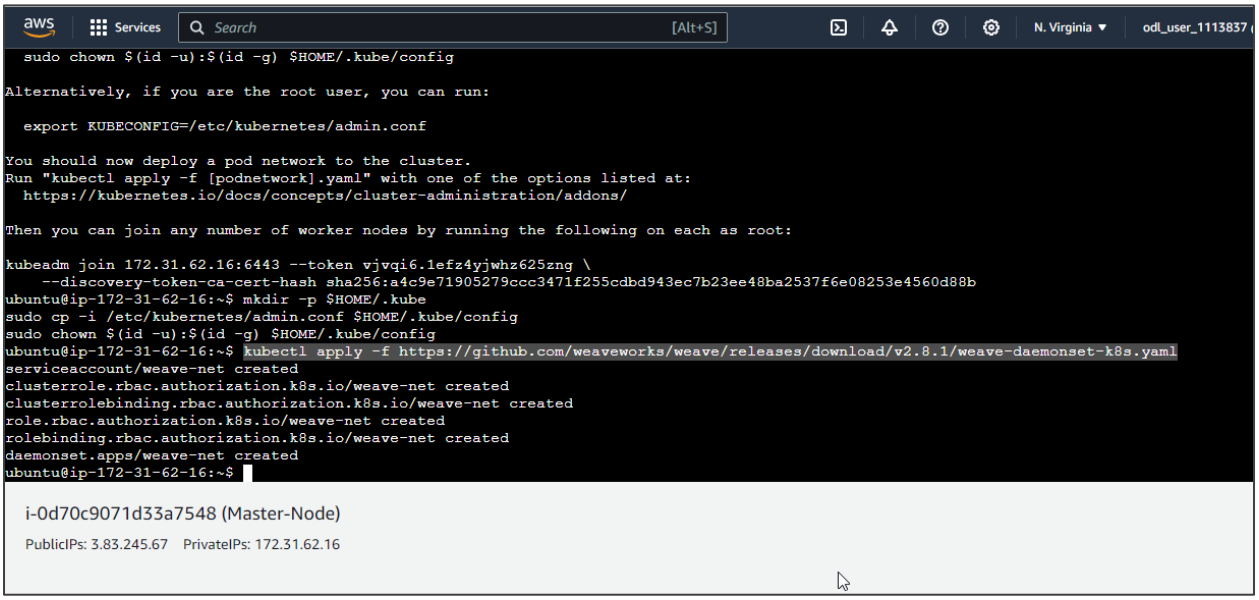


2.2 Execute the following commands to set up local kubeconfig (both for root user and normal user):
mkdir -p $HOME/.kube
sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config
sudo chown $(id -u):$(id -g) $HOME/.kube/config



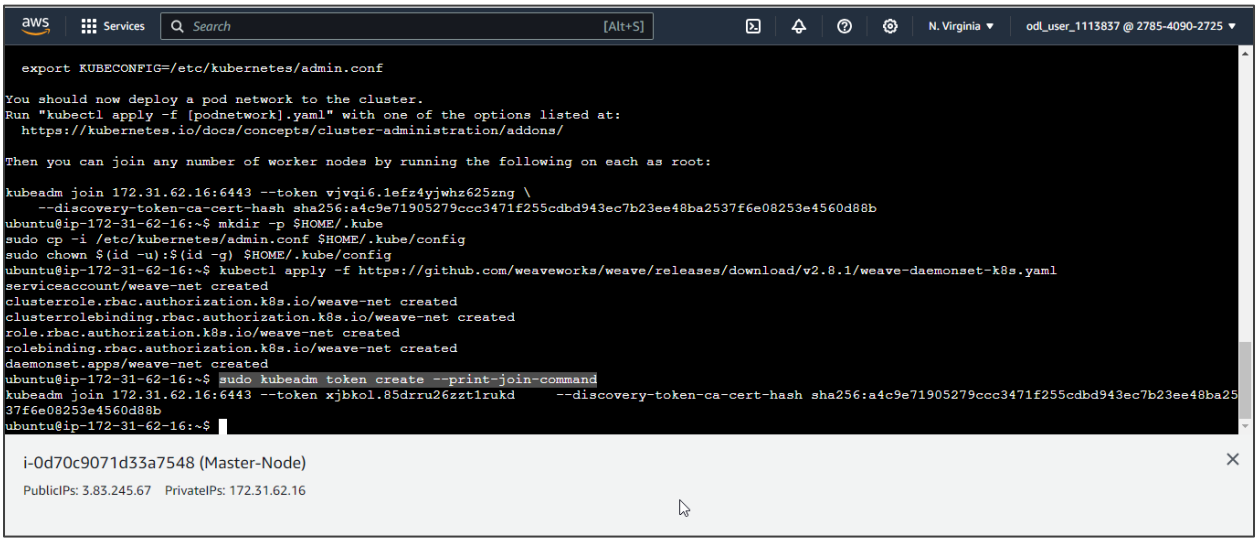
2.3 Run the following command to deploy the weave network:

kubectl apply -f https://github.com/weaveworks/weave/releases/download/v2.8.1/weave-daemonset-k8s.yaml



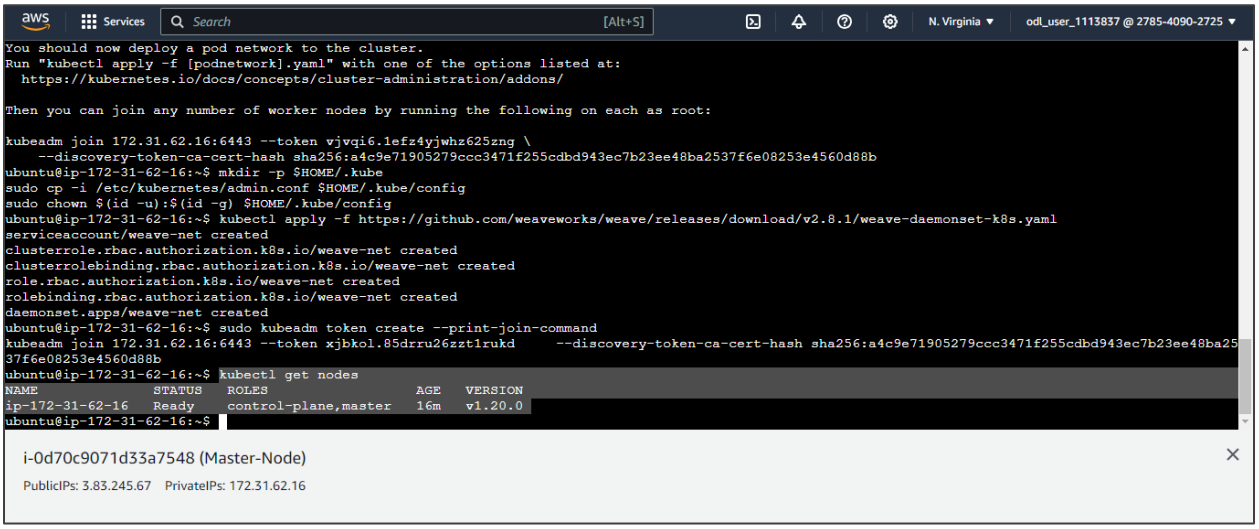
2.4 Run the following command to generate a token for joining the worker node:

sudo kubeadm token create --print-join-command



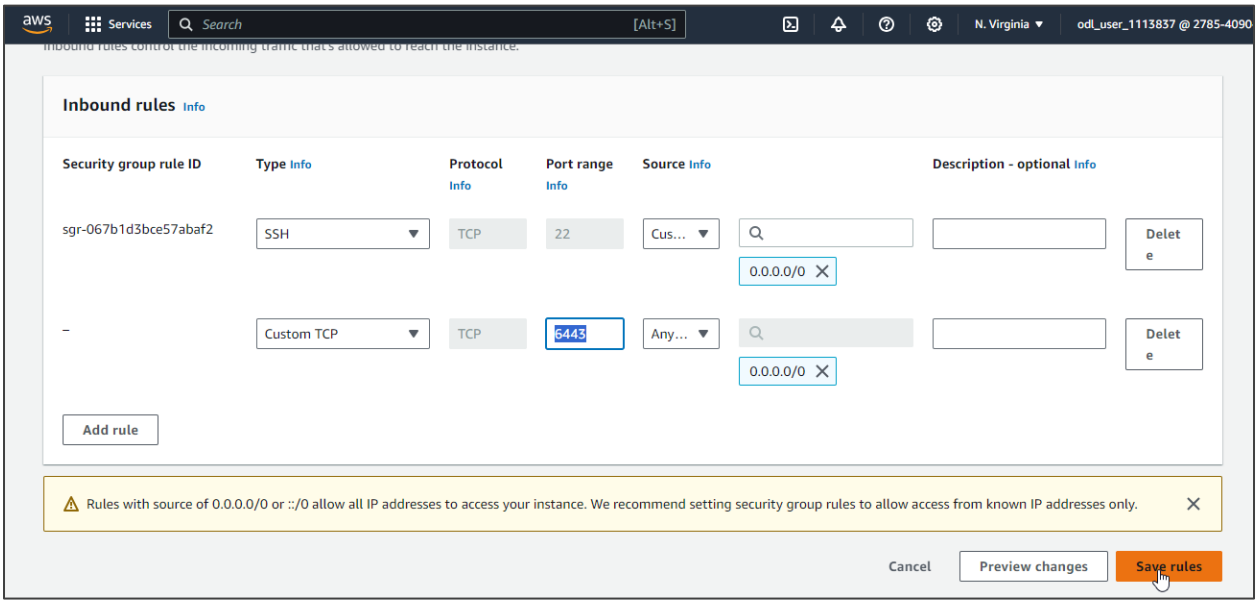
Note: Save the **kubeadm join** command along with the generated token, as it will be required to connect the worker node

2.5 Execute the following command to confirm the node's status:
kubectl get nodes



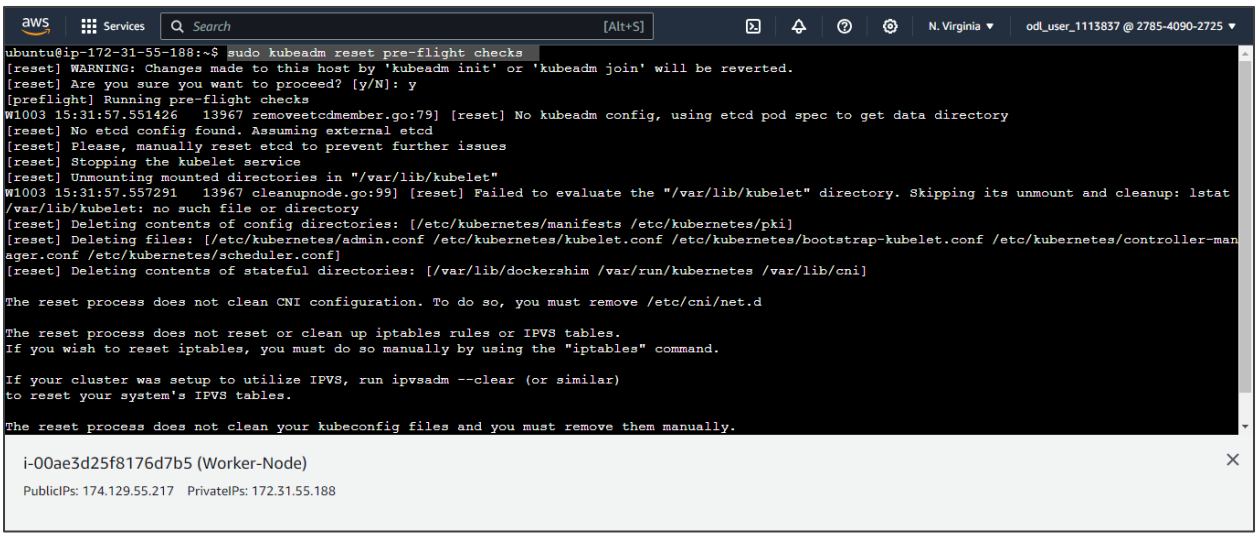
You can see the master node is now ready and operational.

2.6 Open port **6443** in the security group to permit worker nodes to establish connections with the master node

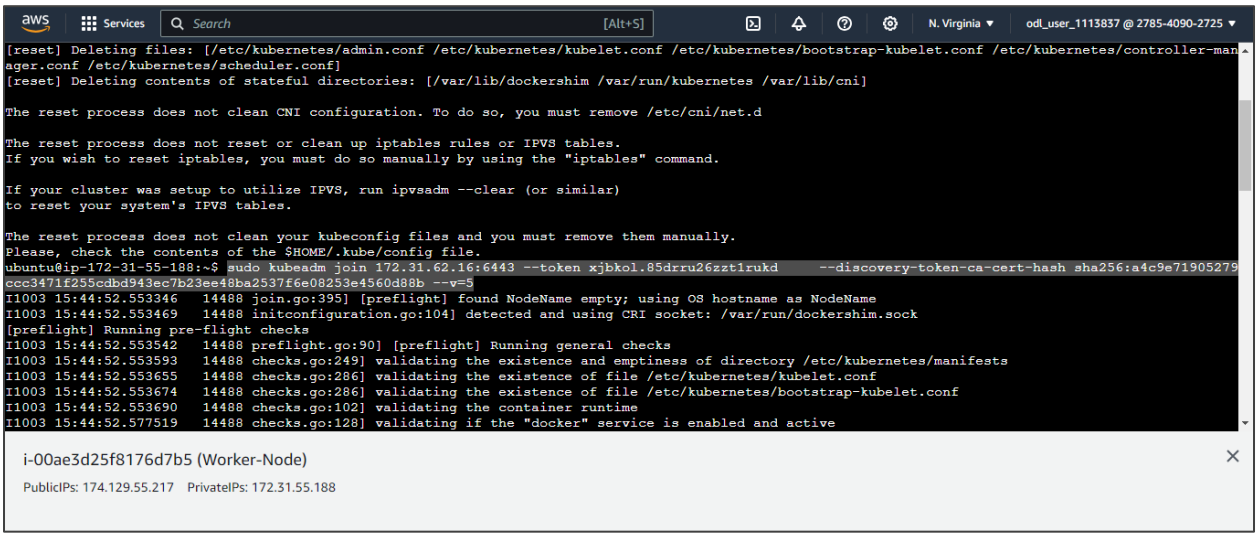


Step 3: Connect the worker instance to the master node

3.1 Execute the below command on the worker node to reset and perform pre-flight checks:
sudo kubeadm reset pre-flight checks

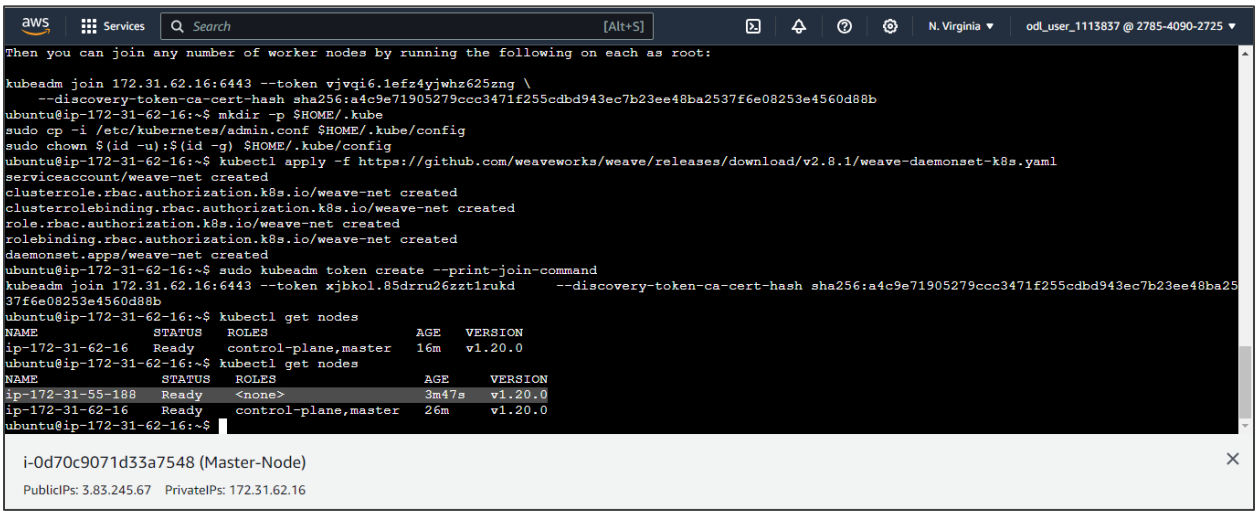


3.2 Paste the **kubeadm join** command you received from the master node and append **--v=5** to the end



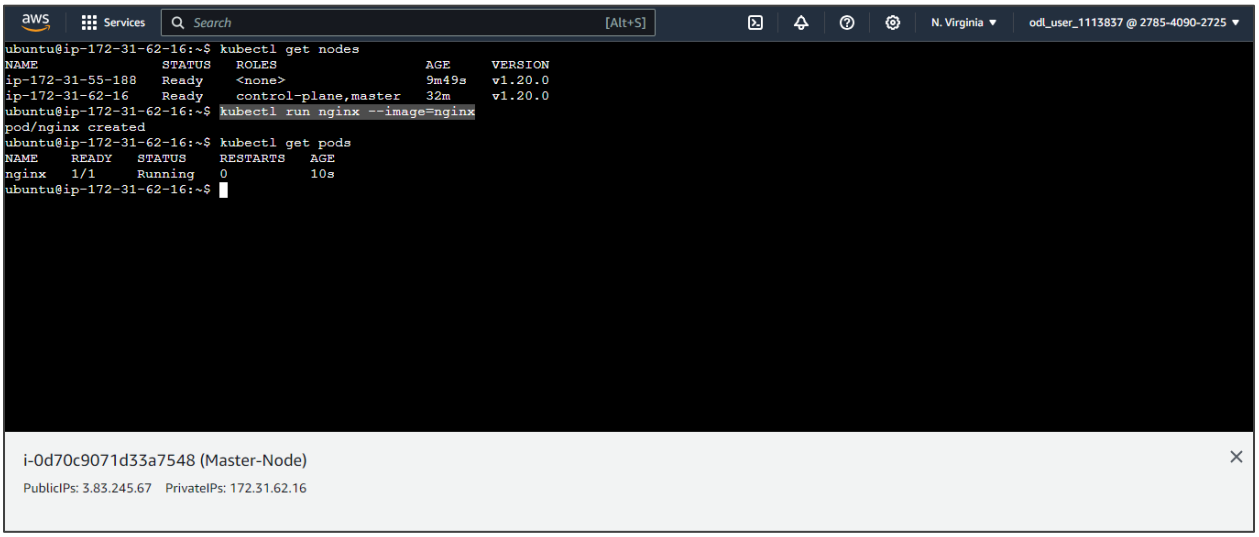
Note: Make sure to use **sudo** before the command

3.3 Return to the master node and run the following command to check if the worker node has successfully joined the cluster:
kubectl get nodes



You can see that the worker node has been successfully integrated into the cluster.

3.4 To deploy a sample pod for testing purposes, execute the given command:
kubectl run nginx --image=nginx



By following these steps, you have successfully established a Kubernetes cluster with a master node and a worker node on EC2 instances.